



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
A.S Nagar, Elupatti– Thanjavur.
 (Approved by AICTE-New Delhi, Affiliated to Anna University, Chennai.)

Department of Mechanical Engineering

CONTINUOUS INTERNAL ASSESSMENT TEST-II, OCT - 2026

Reg.No	8	2	1	9							
Regulation	R2021										
Course Code-Name	OML351 – Introduction to Non-Destructive Testing								SET- A		
Year / Semester	IV & VII								Total Marks:100		
Date - Time	__.10.2026 / 9.30AM TO 12.30PM										
Course Instructor	Dr. S. R. Sathishkumar, AP/ MECH										
C406-OEIII14.3 - Calibrate the instrument and inspect for in-service damage in the components by means of Eddy current testing as well as Thermography testing.											
C406-OEIII14.4 - Differentiate various techniques of UT and AET and select appropriate NDT methods for better evaluation.											
C406-OEIII14.5 - Interpret the results of Radiography testing and also have the ability to analyse the influence of various parameters on the testing.											
L1-Remember , L2-Understand , L3-Apply , L4-Analyze , L5 -Evaluate , L6-Create											

Answer All Questions
PART – A (10×2 = 20 Marks)

Q.NO	QUESTIONS	Marks	CO	LEVEL
1.	List any three fundamental characteristics of eddy currents.	2	C406-OEIII14.3	R
2.	Differentiate between active and passive thermography techniques.	2	C406-OEIII14.3	U
3.	Explain the principle of eddy current testing.	2	C406-OEIII14.3	U
4.	Differentiate between longitudinal waves and shear waves in ultrasonic testing.	2	C406-OEIII14.4	U
5.	Explain the Kaiser effect in acoustic emission testing.	2	C406-OEIII14.4	U
6.	Name two common materials used as couplants in ultrasonic testing.	2	C406-OEIII14.4	R
7.	State the primary function of a filter in film radiography.	2	C406-OEIII14.5	R
8.	List two general methods used for ensuring safety in industrial radiography.	2	C406-OEIII14.5	R
9.	Classify the imaging modalities in radiography based on the detector used.	2	C406-OEIII14.5	U
10.	Compare Computed Radiography (CR) and Direct Radiography (DR) based on their imaging plates.	2	C406-OEIII14.5	U
PART – B (5×13 = 65 Marks)				
11.a	Apply suitable methods of generating the magnetic field in eddy current testing to inspect a component, with neat sketches.	13	C406-OEIII14.3	Ap
OR				
11.b	Demonstrate the instrumentation of eddy current testing with a neat block diagram and apply it to suitable industrial	13	C406-OEIII14.3	Ap

	applications.			
12.a	Apply the laws of thermal imaging (Planck's law, Stefan–Boltzmann law and Wien's displacement law) to explain how a thermography test is carried out.	13	C406-OEIII14.3	Ap
OR				
12.b	Apply the block diagram of an infrared thermography system to a practical inspection scenario and explain the function of each stage.	13	C406-OEIII14.3	Ap
13.a	Apply the principle of the pulse-echo method in ultrasonic testing to detect a defect in a component, with a suitable sketch.	13	C406-OEIII14.4	Ap
OR				
13.b	(i) Demonstrate the characteristics and interpretation of burst (transient) signals in acoustic emission with suitable diagrams. (ii) Demonstrate the significance of continuous signals in acoustic emission with suitable diagrams.	13	C406-OEIII14.4	Ap
14.a	Apply the various stages of radiographic film processing and development, with a flow diagram.	13	C406-OEIII14.5	Ap
OR				
14.b	Apply the four possible interactions between a photon of electromagnetic radiation and the material (photoelectric effect, Rayleigh scattering, Compton scattering and pair production) and explain their practical significance in radiography.	13	C406-OEIII14.5	Ap
15.a	Demonstrate the principle of computed radiography and how it differs from conventional film radiography, with suitable practical applications.	13	C406-OEIII14.5	Ap
OR				
15.b	Apply suitable methods to minimise the factors affecting the quality of radiographic images during testing.	13	C406-OEIII14.5	Ap
PART – C (1×15 = 15 Marks)				
16.a	Evaluate the use of different scan modes (A-scan, B-scan and C-scan) of ultrasonic testing for detecting porosity and cracks in a material, and justify the most suitable choice.	15	C406-OEIII14.4	E
OR				
16.b	Evaluate the different types of ultrasonic probes used in testing and assess how the selection of an appropriate probe influences inspection effectiveness for a given scenario.	15	C406-OEIII14.4	E

Course Instructor

HOD

IQAC Coordinator